

Application Migration Estimation (PERT Model)							
Application	Initial Effort (person-days)	O	M	P	TE Variance Std Dev		
App1	8.00				8.00	0.00	0.00
App2	10.00				10.00	0.00	0.00
App3	6.00				6.00	0.00	0.00
App4	12.00				12.00	0.00	0.00
App5	9.00				9.00	0.00	0.00
App6	15.00				15.00	0.00	0.00
App7	20.00	12.00	20.00	40.00	22.00	21.78	4.67
App8	18.00	12.00	20.00	40.00	22.00	21.78	4.67
App9	22.00	12.00	20.00	40.00	22.00	21.78	4.67
Total	120.00				126.00	65.33	8.08
Management Metrics & Capacity Planning							
Metric	Formula / Basis	Result	Unit	Interpretation / Benchmark	Governance Reference		
Initial total effort	Sum supplied efforts	120.00	person-days	Single-point baseline total across all 9 apps	Baseline Scope		
PERT total effort	Sum of TE for all apps	126.00	person-days	Weighted Expected Duration including App7-9 risk	Risk-Adjusted Scope		
Team size	Given	3.00	engineers	Assigned full-time migration engineering team	Resource Allocation		
Theoretical productive duration	PERT total / team size	42.00	working days	Elapsed project calendar days assuming full concurrency	Schedule Velocity		
Effective working days / month	22 - 5 blackout days	17.00	days/month	Net capacity after recurring month-end freeze window	Operational Constraint		
Blackout-adjusted productive months	Theoretical duration / effective days	2.47	months	Real-world duration factoring migration blackout periods	Delivery Timeline		
Initial effort difference	PERT total - initial total	6.00	person-days	Scope variance introduced by three-point PERT risk model	Effort Variance		
95% planning range for undocumented app	TE +/- 1.96*StdDev	22.00 ± 9.15 days	days	Confidence interval per undocumented app (App7-9)	Statistical Range		
Confidence statement	Planning range only; not a promise	Approx. 95% statistical planning range	statistical guidance	Statistical planning range only; not a delivery promise	Risk Policy		
Model Assumptions & Governance Framework							
Item / Category	Assumption / Specification	Value / Formula	Impact Area	Governance Rationale	Status / Source		
App1–App6 estimates	Use supplied single-point effort figures exactly as given	8, 10, 6, 12, 9, 15 person-days	Documented Workloads	Verified architecture documentation; zero variance modeled	Project Specification		
App7–App9 PERT	Three-point estimation for undocumented applications	O=12, M=20, P=40 person-days	Undocumented Workloads	Uncertainty modeled due to undocumented legacy code debt	SME Consensus		
PERT Calculation Formulas	Weighted expected duration & statistical dispersion	TE=(O+4M+P)/6 ; Var=((P-O)/6)^2	Schedule Estimation	Standard beta distribution weighting for schedule modeling	PMI / PERT Standard		
Engineering Team Capacity	Dedicated migration engineering staff	3 full-time engineers	Resource Allocation	Parallel execution assuming 100% active concurrency	Given Constraint		
Working Calendar & Blackout Window	Gross monthly working days and change-freeze blackouts	22 days gross - 5 blackout days = 17 effective days/month	Timeline Planning	Final 5 business days reserved for enterprise financial close freeze	Enterprise Policy		
Statistical Confidence Statement	95% statistical planning range specification	TE ± 1.96 * StdDev (Planning range only; not a promise)	Risk Management	Guidance range for executive planning; not a delivery promise	Governance Mandate		